EXACT RATIONAL SIGN-CELL EXCLUSION ON THE BALANCED $(2,2,2,2)$ HEILBRONN CONFIGURATION SPACE

ARGUS OPEN RESEARCH PROJECT

Abstract. For the Heilbronn eight-point triangle problem in the unit square, a proposed counterexample to the exact maximum minimum area $1/9$ requires an eight-point configuration in $[0,1]^2$ whose $\binom{8}{3} = 56$ determined triangles all have area strictly exceeding $1/9$. We establish an exact rational sign-cell exclusion of the 175-leaf root-118 balanced $(2,2,2,2)$ corner regime, proving that every cell in that partition violates at least one triangle-area inequality.

1. Main Scoped Exclusion

Theorem 1.1. *In the one-corner balanced* $(2,2,2,2)$ *parameterization of eight points in* $[0,1]^2$*, the* 175 *rational box leaves of the root-*118 *frontier admit exact positive rational linear combinations of the* 56 *signed determinant constraints* $2\,\mathrm{Area}(p_i,p_j,p_k) > 2/9$ *that are strictly negative throughout each cell, excluding a strict* $> 1/9$ *configuration across that entire region.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1380 in the domain of discrete geometry follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped exclusion certificate* deciding the scope: exhaustive rational exclusion of all 175 manifest leaves of the root-118 balanced (2,2,2,2) eight-point Heilbronn frontier.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- unrestricted eight-point Heilbronn configurations.

References

[1] Primary problem list and source literature for Result 1380 (discrete geometry), archived with exact theorem locators and verification certificates in the accompanying repository package.

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